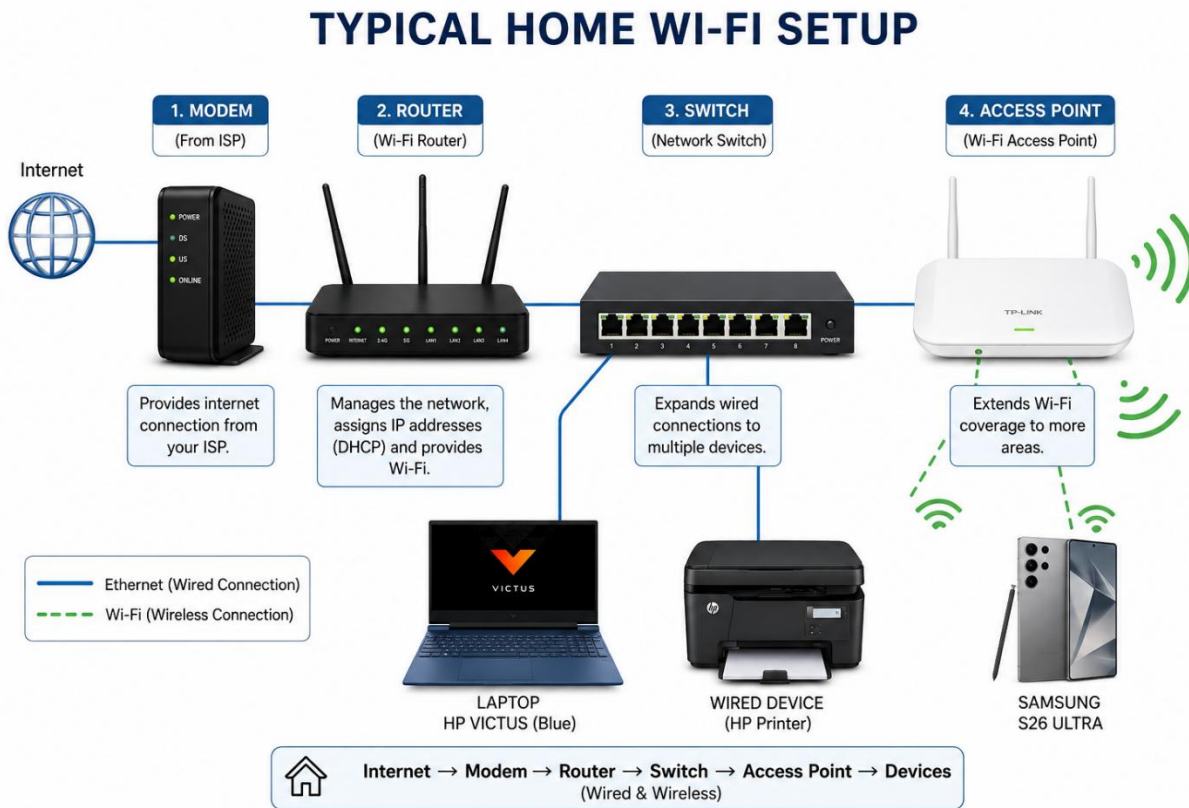


Assignment Day-1

1. Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.



2. List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').



Networking Device	Main Function	Real-Life Example
Modem	Connects your home network to the internet service provided by your ISP.	Used at home to receive internet from Jio, Airtel, or BSNL .
Router	Directs network traffic between devices and shares the internet connection. It can also provide Wi-Fi.	Used at home to connect your laptop, smartphone, and smart TV to the internet.
Switch	Connects multiple wired devices on the same local network and allows them to communicate.	Used in an office to connect computers, printers, and servers using Ethernet cables.
Access Point (AP)	Provides or extends Wi-Fi coverage so wireless devices can connect to a network.	Used in a college, hotel, or large house to provide Wi-Fi in areas where the router's signal is weak.
Network Cable (Ethernet Cable)	Carries data between networking devices and provides a wired network connection.	Used to connect a router to a PC, switch, printer, or access point .

3. Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.



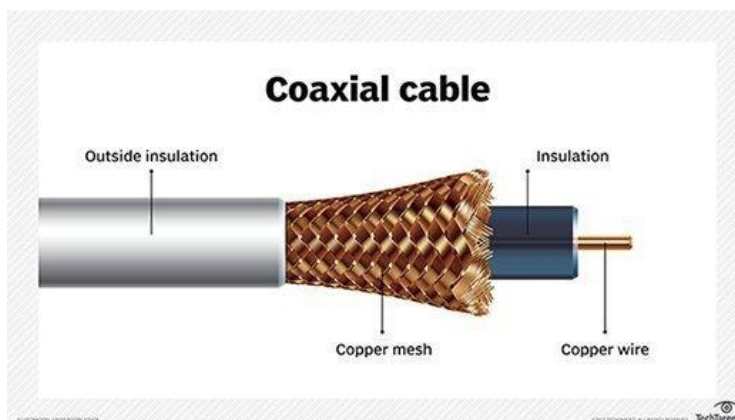
1. **Ethernet Cable(RJ45)**

RJ45 Connector

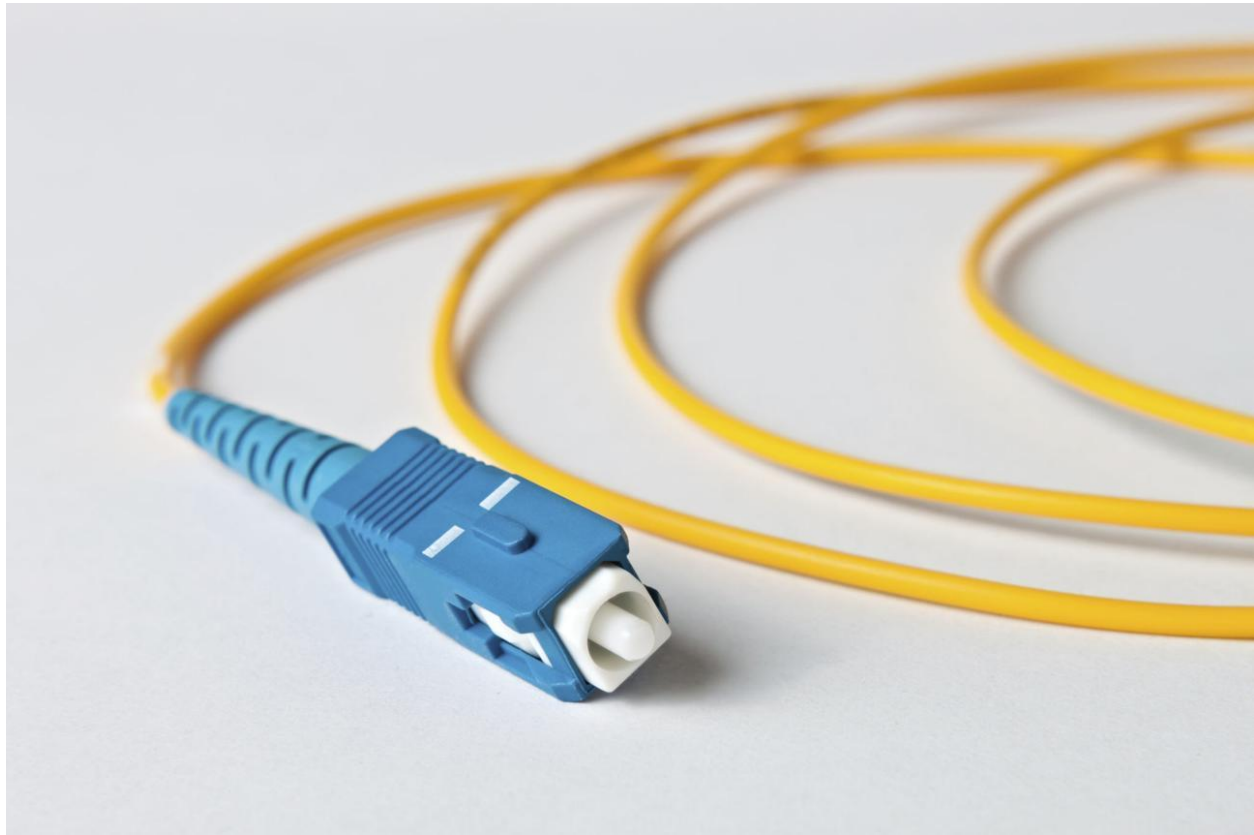


TechTerms.com

2. **Coaxial Cable**



3. Fiber Optic Cable



No.	Cable Type	Common Use
1	Ethernet Cable	Commonly used to connect computers, routers, switches, printers, and access points in home and office networks.
2	Coaxial Cable	Commonly used for cable TV, cable internet, satellite TV, and some CCTV systems .
3	Fiber Optic Cable	Commonly used for high-speed internet, data centers, telecom networks, and long-distance communication .

4. Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.



Feature	Router	Switch
Main Use	Connects different networks and shares the internet.	Connects multiple devices within the same network.
Typical Location	Usually found at home, office, or ISP network.	Usually found in offices, schools, data centers, and server rooms.
How It Handles Data	Uses IP addresses to decide where packets should go.	Mainly uses MAC addresses to forward frames to the correct device.
Internet Connection	Can connect a local network to the Internet.	Normally does not directly connect a LAN to the Internet.
Devices Connected	Connects different networks and devices such as PCs, phones, and access points.	Connects multiple wired devices such as PCs, printers, and servers.
Example	A home Wi-Fi router sharing internet with a laptop and smartphone.	An office switch connecting 20 computers and printers together.